

$$\ddot{\Theta}_i + \mathcal{H}_i \Theta_i = (\bar{u}_i, \bar{Q})$$

$$\Theta_i = \Theta_i^{\text{огн}} + \Theta_i^{\text{расм.}}$$

$$c_i \sin(\sqrt{\mathcal{H}_i} t + \alpha_i)$$

$$\} Q_i - \text{непрерывн. } \omega T = \frac{2\pi}{\mathcal{H}^*}$$

$$\Theta_i = \sum_{k=0}^{\infty} b_{ik} \sin(k \mathcal{H}^* t + \alpha_k)$$

в прошлой лекции ошибка, сумм. беск !

$$\Theta_i^{\text{расм.}} = \sum_{k=0}^{\infty} b_{ik}^* \sin(k \mathcal{H}^* t + \alpha_k)$$

$$\Theta_i^{\text{расм.}} = \sum_{k=0}^{\infty} \frac{b_{ik}}{\mathcal{H}_i - k^2 (\mathcal{H}^*)^2} \sin(k \mathcal{H}^* t + \alpha_k)$$

$$\leftarrow \mathcal{H}_i \neq k^2 (\mathcal{H}^*)^2$$

$$\Theta_i = c_i \sin(\sqrt{\mathcal{H}_i} t + \alpha_i) + \Theta_i^{\text{расм.}}$$

$$\bar{q} = \sum \bar{u}_i \cdot \Theta_i = \sum_{i=1}^n c_i \bar{u}_i \sin(\sqrt{\mathcal{H}_i} t + \alpha_i) + \sum_{i=0}^n \bar{u}_i \Theta_i^{\text{расм.}}$$

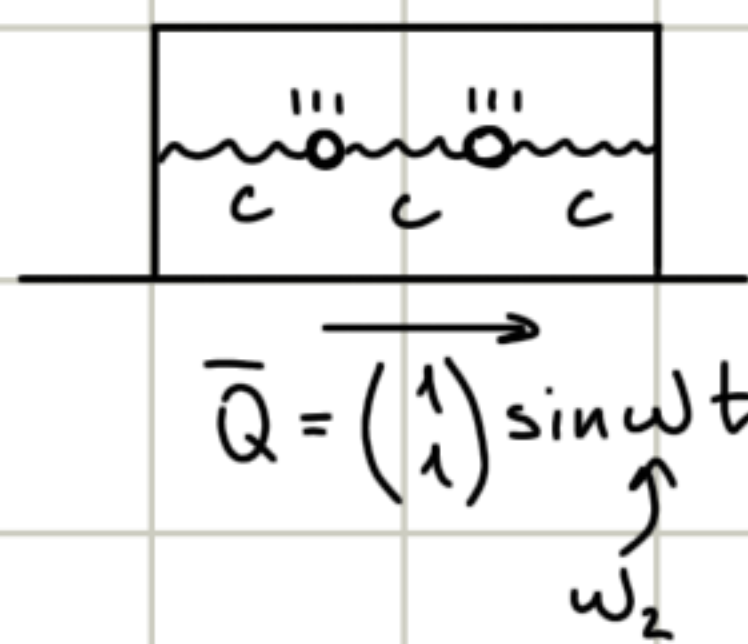
$$\{ A \ddot{\bar{q}} + C \bar{q} = \bar{Q} \}$$

$$\text{если } \mathcal{H}_i = k^2 (\mathcal{H}^*)^2 ?$$

$$1) b_{ik} = 0, \text{ но все ок}$$

$$\Theta_i = b_{ik} \sin(k \mathcal{H}^* t + \alpha_k)$$

$$\Theta_i = 0, \text{ если } (\bar{u}_i, \bar{Q}) = 0 !$$



$$\bar{Q} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \sin \omega t$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}_{\omega_1} = \bar{u}_1$$

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}_{\omega_2} = \bar{u}_2$$

$$\ddot{\Theta}_1 + \omega_1^2 \Theta_1 = (\bar{u}_1, \bar{Q})$$

$$\ddot{\Theta}_2 + \omega_2^2 \Theta_2 = (\bar{u}_2, \bar{Q})$$

$$\left\{ \begin{array}{l} \ddot{\Theta}_1 + \omega_1^2 \Theta_1 = 2 \sin \omega t \\ \ddot{\Theta}_2 + \omega_2^2 \Theta_2 = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} \ddot{\Theta}_1 + \omega_1^2 \Theta_1 = 2 \sin \omega t \\ \ddot{\Theta}_2 + \omega_2^2 \Theta_2 = 0 \end{array} \right.$$

$$2) b_{ik} \neq 0$$

$$\ddot{\Theta} + \omega^2 \Theta = a \sin \omega t$$

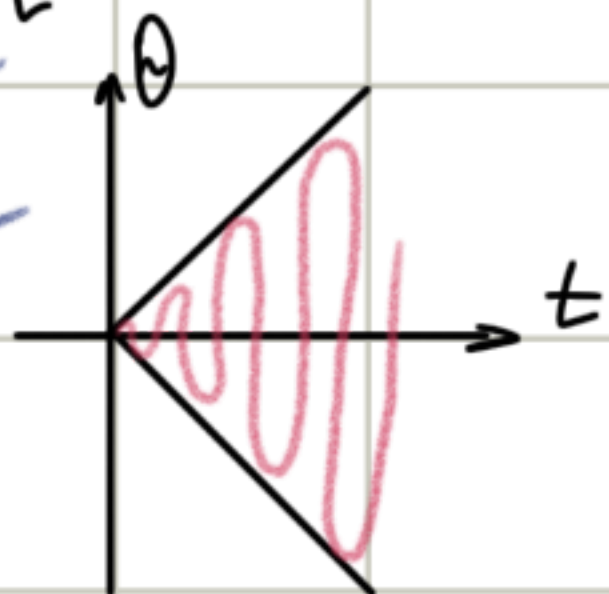
$$\Theta = c \sin(\omega t + \alpha) + \Theta^{\text{расм.}}$$

$$\Theta^{\text{расм.}} = b t \cos \omega t$$



$$\theta^{\text{наст.}} = - \frac{a}{2\omega} t \cos \omega t$$

резонанс



$$\bar{q} = \sum_{i=1}^n \bar{u}_i \theta_i$$

Если  $\omega \in \{\omega_i\} \Rightarrow$  резонанса может не быть (1)

Если резонанс, то  $\omega \in \{\omega_i\}$

### Вынужденные колебания диссип. сист.

1)  $\exists$  есть  $\Pi$ -компон. эн.

2)  $\exists \bar{Q}(\bar{q}, \dot{\bar{q}}, t)$  - обобщ. сила для некон. сил ( $\exists$  кон. диссипация)

$$Q_i = - \frac{\partial K}{\partial \dot{q}_i}$$

$$K = \sum_{k=1}^n b_{ik} \dot{q}_i \dot{q}_k$$

ф-я Релея

$$Q_i = - \sum_{k=1}^n b_{ik} \dot{q}_k$$

$$b_{ik} = b_{ki}$$

3)  $\exists \bar{Q}(t)$  - внеш. сила

$\bar{Q}(t)$  грим. мала, чтобы осм. в окр. ПР

$$T = \frac{1}{2} \sum_{i,k} a_{ik} \dot{q}_i \dot{q}_k$$

$$\Pi = \frac{1}{2} \sum_{i,k} c_{ik} q_i q_k$$

$$A\ddot{\bar{q}} + B\dot{\bar{q}} + C\bar{q} = \bar{Q}(t)$$

$$Q_i = - \sum_k b_{ik} \dot{q}_k$$

$$\bar{q} = \bar{q}^{\text{огн.}} + \bar{q}^{\text{наст.}}$$

$$\bar{q}^{\text{огн.}} : A\ddot{\bar{q}} + B\dot{\bar{q}} + C\bar{q} = 0$$

$$\bar{q} = \bar{u} e^{\lambda t}$$

$$\det(A\lambda^2 + B\lambda + C) = 0$$

$$\Rightarrow \{\lambda_i\}$$

$$(A\lambda^2 + B\lambda + C)\bar{u} = 0 \Rightarrow \bar{u}$$

Если ос. угн., то  $\text{Re } \lambda < 0$  !

предполож. про  $\bar{Q}(t)$

$$1) \bar{Q}(t) = \begin{pmatrix} Q_1 \\ \vdots \\ Q_n \end{pmatrix} = \begin{pmatrix} Q_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ Q_2 \\ \vdots \\ 0 \end{pmatrix} + \dots + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ Q_n \end{pmatrix}$$

$$2) Q_e = h_e \sin(\lambda t)$$

$$3) \sin \rightarrow e$$

$$Q_e = h_e e^{i\lambda t}$$

$$A\ddot{\bar{q}} + B\dot{\bar{q}} + C\bar{q} = \begin{cases} h_e e^{i\lambda t}, & k=l \\ 0, & k \neq l \end{cases}$$



$$\bar{q} = \bar{D} e^{i\omega t} *$$

$$(-A\omega^2 + B(i\omega) + C)\bar{D}e = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \cdot h_e$$

$H\bar{x} = b$  правило Крамера

$$\Delta = \det(-A\omega^2 + B(i\omega) + C)$$

$$\bar{D}_{ek} = \frac{\Delta_{ek}(i\omega)}{\Delta(i\omega)} h_e$$

$\Delta_{ek}(i\omega)$  - алгебраич. дополн. к  $\{l, k\}$  элементу в матриц. \*

$$\begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

$$-A\omega^2 + B(i\omega) + C$$

$$w_{ek} = \frac{\Delta_{ek}(i\omega)}{\Delta(i\omega)} - \text{амплитудно-фаз. хар-ка}$$

(частотная хар-ка)

1) при  $\omega \rightarrow \infty \quad |w_{ek}| \rightarrow 0$

2)  $w_{ek} = \underbrace{\operatorname{Re} w_{ek}}_{\text{ампл.}} + i \underbrace{\operatorname{Im} w_{ek}}_{\text{фаз.}}$

$$w_{ek}(\omega=0) = \operatorname{Re} w_{ek}(0)$$

логарифм АФХ



$$\operatorname{tg} \psi = \frac{\operatorname{Im} w_{ek}(\infty)}{\operatorname{Re} w_{ek}(\infty)} = \begin{cases} \rightarrow 0 \\ \rightarrow \infty \end{cases}$$

$$3) w_{ek}(i\omega) = \underbrace{|w_{ek}(i\omega)|}_{\substack{\operatorname{Re} w_{ek}(i\omega) \\ \text{ампл. хар.}}} e^{i \underbrace{\arg w_{ek}(i\omega)}_{\substack{\psi_{ek}(i\omega) \\ \text{фаз. хар-ка}}}}$$

$$\bar{D}_{ek} = w_{ek} \cdot h_e = |w_{ek}(i\omega)| h_e e^{i \arg w_{ek}(i\omega)}$$

$$q_k = |w_{ek}(i\omega)| h_e e^{i(\omega t + \arg w_{ek}(i\omega))}$$

физ. ответ АФХ

Пронзв.  $Q_e$  на частоте  $p$  с ампл.  $A$

$$q_k = A \cdot |w_{ek}(ip)| e^{i(p t + \arg w_{ek}(ip))}$$

прим.

$$a\ddot{q} + b\dot{q} + cq = h e^{i\omega t}$$



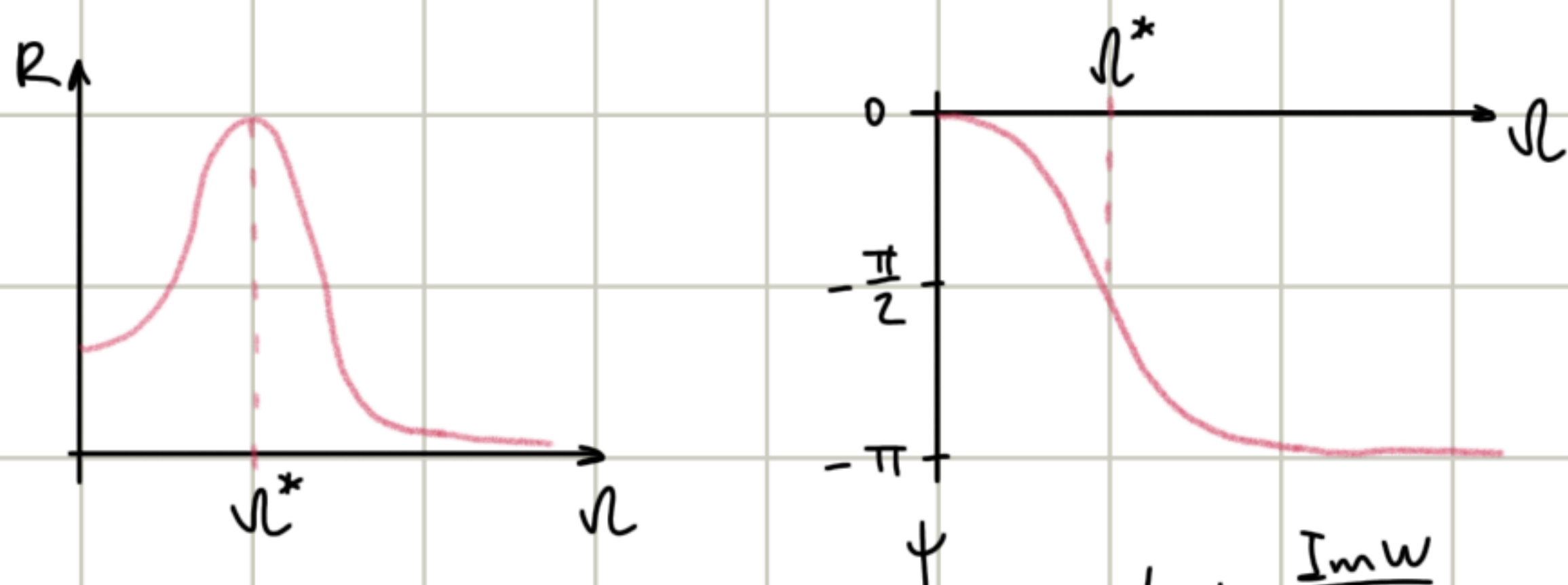
$$q = D e^{i\omega t}$$

$$(-a\omega^2 + b(i\omega) + c) \cdot D = h$$

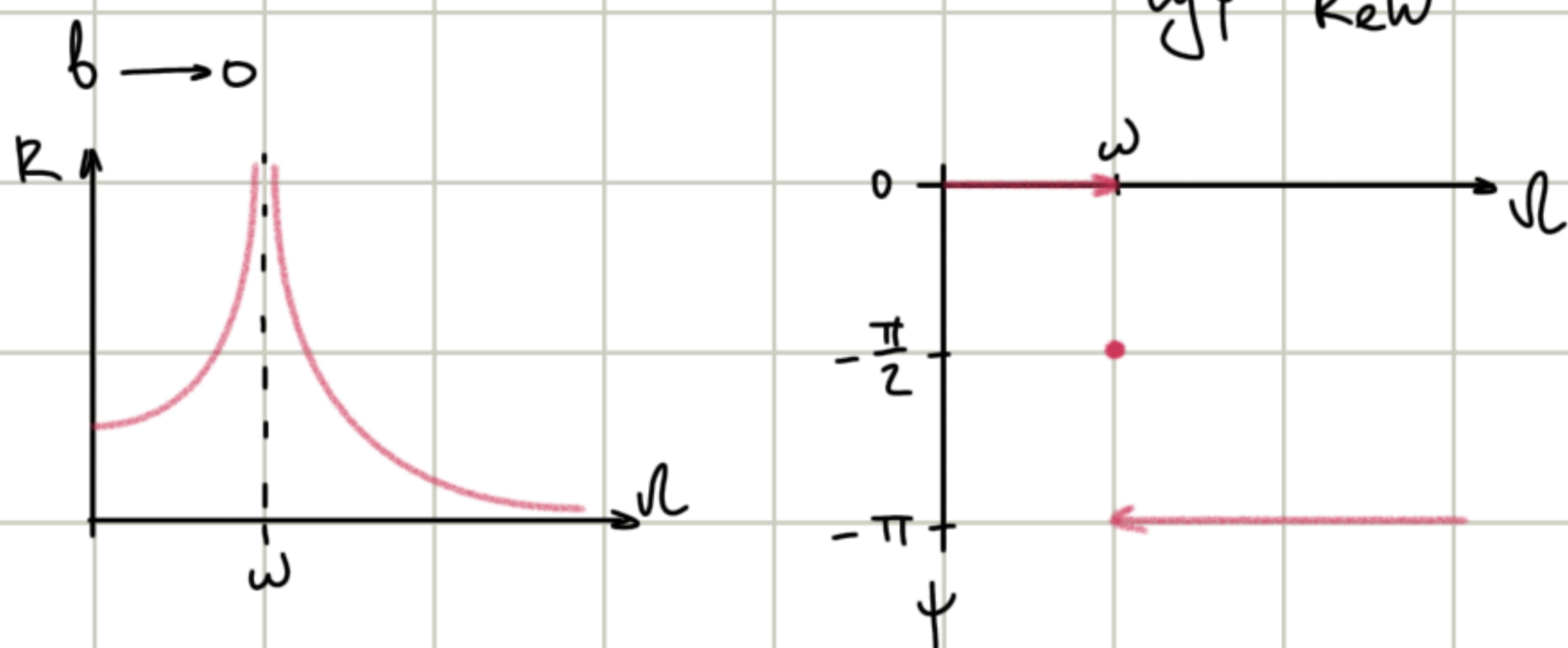
$$D = \frac{h}{-a\omega^2 + b(i\omega) + c}$$

$$w = \frac{1}{-a\omega^2 + b(i\omega) + c}$$

$$R = |w| \quad \omega^* = \sqrt{\frac{c}{a}}$$



$$\operatorname{tg} \psi = \frac{\operatorname{Im} w}{\operatorname{Re} w}$$



## Лагранжева механика

### Ур-е Лагранжа

Лагр. мех.:  $q, \dot{q}, t$   $L$

Лам. мех.:  $q, p, t$   $H$

$$\begin{cases} \dot{q} = \bar{Q}(q, p, t) \\ \dot{p} = \bar{P}(q, p, t) \end{cases}$$

- лагранжева, если ЭН:

$$Q_i = \frac{\partial H}{\partial p_i} \quad , \quad p_i = - \frac{\partial H}{\partial q_i}$$

критерий лам.:

$$\frac{\partial Q_i}{\partial p_j} = \frac{\partial Q_j}{\partial p_i} \quad \left( \frac{\partial^2 H}{\partial p_i \partial p_j} \right)$$

$$\frac{\partial P_i}{\partial q_j} = \frac{\partial P_j}{\partial q_i} \quad \left( \frac{\partial^2 H}{\partial q_i \partial q_j} \right)$$

$$\frac{\partial Q_i}{\partial q_j} = - \frac{\partial P_j}{\partial p_i} \quad \left( \frac{\partial^2 H}{\partial p_i \partial q_j} \right)$$

II.  $p_i$  - обобщ. импульсы

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = f_i(\bar{q}, \dot{\bar{q}}, t)$$



$$\Rightarrow \dot{q} = \dot{q}(q, p, t) \quad \left| \frac{\partial \vec{f}}{\partial \dot{q}} \right| \neq 0 \Rightarrow$$

$$\text{wenn } \left| \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \right| \neq 0, \text{ so } \dot{q} = \dot{q}(q, p, t)$$

$$\mathcal{L} = T - \Pi \quad (T - V), \quad T = T_2 + T_1 + T_0$$

$$\left| \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \right| = \left| \frac{\partial^2 T_2}{\partial \dot{q}_i \partial \dot{q}_j} \right| = \|a_{ik}\| \neq 0$$